Integrated E-Commerce Sales and Marketing Analytic

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Course Name: AI-Driven Data Analytics

Batch No: FNB16

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1. Project Overview and Objective

Project Overview

- ➢ Analyzed e-commerce sales, inventory, customer, product, returns, and marketing campaign data.
- ➢ Performed data cleaning, preprocessing, and analysis using Microsoft Excel.
- ➢ Built a structured dataset for Power BI reporting.
- ➢ Created business-ready data for interactive dashboard development.

Project Objectives

- ➢ Clean and organize raw business data.
- ➢ Improve data quality through preprocessing.
- ➢ Prepare datasets for business intelligence reporting.
- ➢ Analyze sales, inventory, customer, and marketing performance.
- ➢ Develop interactive dashboards using Power BI.
- ➢ Generate meaningful business insights for decision-making.

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2. Data Sources

- ➢ **Source:**https://excelx.com/practice-data/generators/ecommerce-marketing/
- ➢ TIME LINE: 2022-2026
- ➢ **Domain**: Commerce & Marketing Campaign

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3. Problem Statement

- ➢ Analyze overall e-commerce business performance.
- ➢ Evaluate sales performance across products and categories.
- ➢ Monitor customer purchasing behavior.
- ➢ Analyze inventory availability across warehouses.
- ➢ Identify product return patterns.
- ➢ Evaluate marketing campaign effectiveness.
- ➢ Create an interactive business intelligence dashboard.

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4. Attribute (Column /Features) Details:

Field	**Type**	**Details**
CustomerID	Text	Unique customer identifier.
Customer Name	Text	Customer’s full name.
Country	Text	Customer’s country or region.
OrderID	Text	Unique order identifier.
TotalAmount	Currency	Total order value.
Order Date	Date	Order placement date.
Order Status	Text	Current status of the order.
Product Id	Text	Unique product identifier.
Quantity	Whole Number	Units ordered.

5.Tools & Technologies

- **Excel:** Data cleaning, Transformation, Sorting & Filtering, Formatting, Conditional Formatting, Power Query, Pivot Table.
- **Power BI**: Data Modeling, DAX, Calculated Column, Calculated Measures, Visualization.

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Cleaning & Transformation:** Removed duplicates, handled missing values, standardized formats, and created calculated fields.
- **Filtering & Sorting:** Organized data to focus on relevant records.
- **Pivot Tables:** Generated Pivot Tables for data summarisation and initial insights. ➢ Fact and Dimension Table

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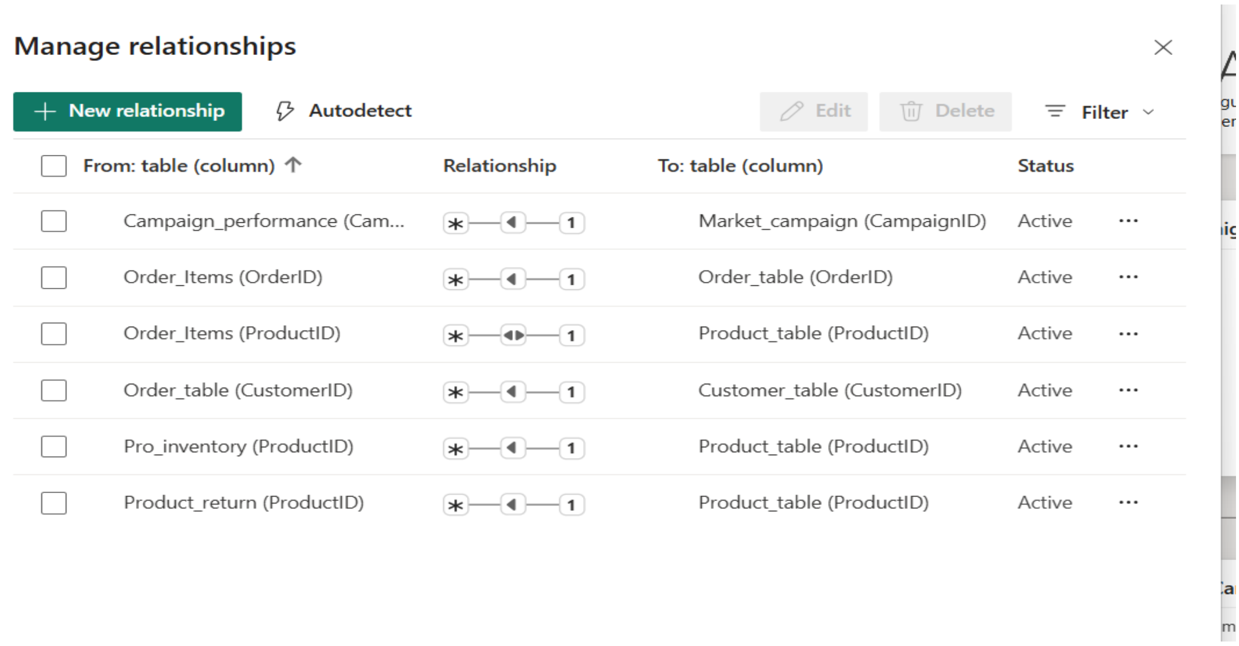
7. Data Modelling and DAX (Power BI)

- Designed a structured data model in Power BI for efficient analysis and reporting.
- Organized the dataset into fact tables and dimension tables.
- Established relationships between tables using common key fields such as CustomerID, ProductID, and OrderID.
- Followed a star schema approach to improve dashboard performance and data filtering.
- Used DAX formulas to create calculated columns and measures.
- Created key performance indicators such as Total Sales, Total Orders, Average Order Value, and Customer Count.
- Calculated return-related metrics such as Return Rate and returned order count.
- Measured marketing performance using campaign-related DAX measures.
- Improved accuracy and consistency of business insights through proper data modelling and DAX calculations.

Relationship and Cardinality

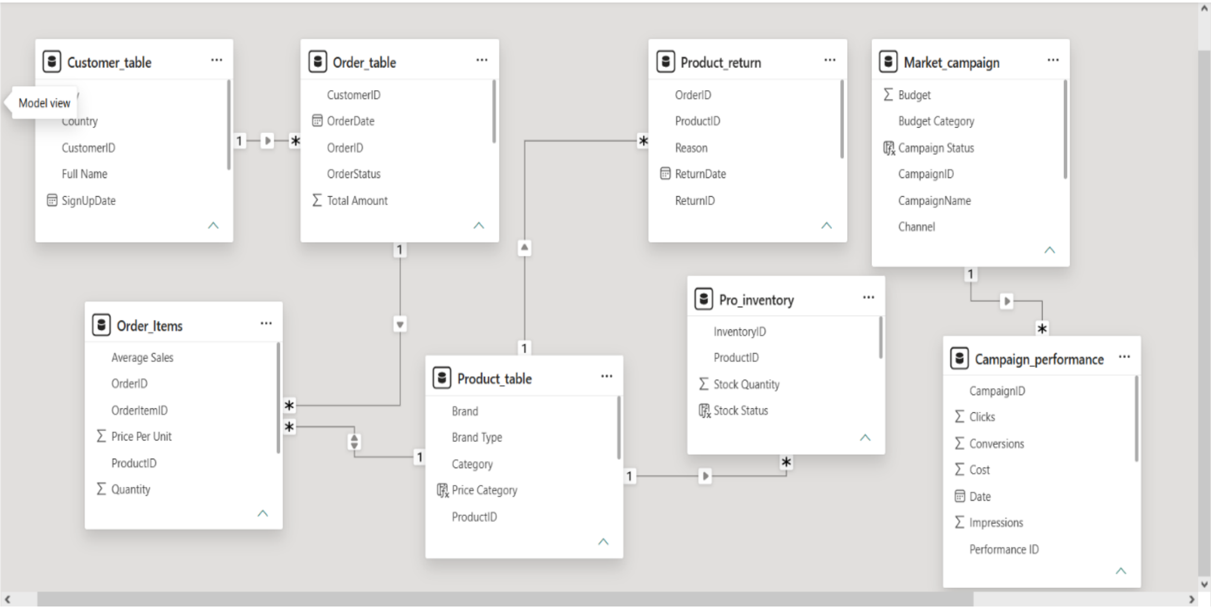
- Created relationships between fact and dimension tables using common key columns.
- Used **CustomerID** to connect the **Customer Table** with the main fact/order-related table.
- Used **ProductID** to connect the **Product Table** with the order item, inventory, and return tables.
- Used **OrderID** to connect the **Order Table** with the **Order Item Table** and **Product Return Table**.

- Used **CampaignID** to connect the **Marketing Campaign Table** with the **Campaign Performance Table**.



Data Modeling

- The **Order Item Table** was used as the central fact table because it contains transaction-level sales details.
- The fact table stores measurable values such as quantity, sales amount, and order-related metrics.
- The **Customer Table** was used as a dimension table to analyze customer details and purchasing behavior.
- The **Product Table** was used as a dimension table to analyze product category, product name, and product performance.
- The **Order Table** was connected to the fact table to provide order-level information such as order date, order status, and payment details.
- The **Product Inventory Table** was used to monitor stock availability and warehouse-level inventory details.
- The **Product Return Table** was used to analyze returned products, return quantity, return reason, and return patterns.
- The **Marketing Campaign Table** and **Campaign Performance Table** were used to evaluate campaign effectiveness and marketing results.
- Relationships were created using key fields such as **CustomerID**, **ProductID**, **OrderID**, and **CampaignID**.
- A star schema approach was followed to improve dashboard performance and simplify report filtering.
- One-to-many relationships were used between dimension tables and the fact table.
- This data model helped create accurate KPIs, interactive visuals, and business insights in Power BI.



Calculated Columns & DAX Measures:

Created New Table: Date Table
Formula: Date Table =
ADDCOLUMNS (
    CALENDAR (DATE(2023,1,1), DATE(2026,12,31)),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month Name", FORMAT([Date], "MMMM"),
    "Quarter", "Q" & FORMAT([Date], "Q"),
    "Day", DAY([Date]),
    "Day Name", FORMAT([Date], "DDDD"),
    "Week Number", WEEKNUM([Date])
)

Calculated Column-
Table Name: Marketing Campaign -Column Named As:Campaign Status
Formula: Campaign Status =
IF(
    Market_campaign[EndDate]< TODAY(),
    "Completed",
    "Active"
)

Calculated Column
Table Name; Product Inventory-Column Named As: Stock Status
Formula: Stock Status = SWITCH(
    TRUE(),
    Pro_inventory[Stock Quantity] = 0, "Out of Stock",

```
    Pro_inventory[Stock Quantity] < 70, "Low Stock",
    Pro_inventory[Stock Quantity] < 250, "Medium Stock",
    "High Stock"
)
```

Calculated Column
Table Name: Product Return
Formula: Return Month = FORMAT(Product_return[ReturnDate], "MMM YYYY")

Calculated Column
Table Name: Product Table : Average Price = ₹ 4,711.61

Formula:
```
Price Category = SWITCH(
    TRUE(),
    Product_table[Unit Price] < 2000, "Low",
    Product_table[Unit Price] < 5000, "Average",
    "High"
)
```

Calculated Column
Table Name: Customer Table
Formula:
```
City Category =
SWITCH(
    TRUE(),
    'Customer_table'[City] IN
{"Mumbai","Delhi","Chennai","Bangalore","Kolkata","Hyderabad","Pune"}, "Metro City",
    'Customer_table'[City] IN {"Jaipur","Ahmedabad","Surat"}, "Tier-2 City",
    "Other City"
)
```

Dax : Calculated Measures

```
CTR % =
DIVIDE(
SUM(Campaign_performance[Clicks]),SUM(Campaign_performance[Impressions]),0)
```

```
Conversion Rate % =
DIVIDE(
    SUM(Campaign_performance[Conversions]), SUM(Campaign_performance[Clicks]),0)
```

Cost Per Click =

```
DIVIDE(
   SUM(Campaign_performance[Cost]), SUM('Campaign_performance'[Clicks]),0)
```

```
Total Customers = COUNTROWS(Customer_table)
```

```
Processing Orders =
CALCULATE(COUNTROWS(Order_table),Order_table[OrderStatus]="Processing")
```

```
Delivered Orders =
CALCULATE(COUNTROWS(Order_table),Order_table[OrderStatus]="DELIVERED")
```

```
Return Orders = COUNTROWS(Product_return)
```

```
Products Counts By Brand = COUNTROWS(Product_table)
```

```
Total Products = DISTINCTCOUNT(Product_table[ProductID])
```

```
Avg Unit Price = AVERAGE(Product_table[Unit Price])
```

```
Max Stock = MAX(Pro_inventory[Stock Quantity])
```

```
StockByWarehouse = SUM(Pro_inventory[Stock Quantity])
```

```
Total Stock = SUM(Pro_inventory[Stock Quantity])
```

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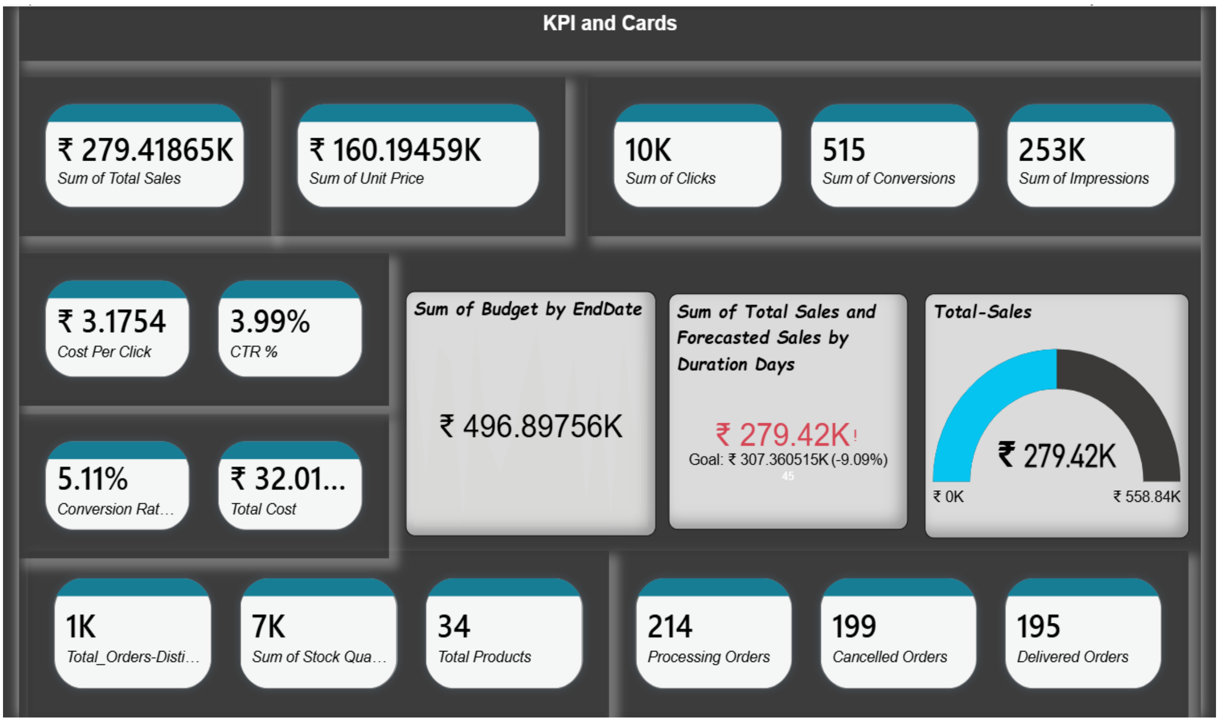
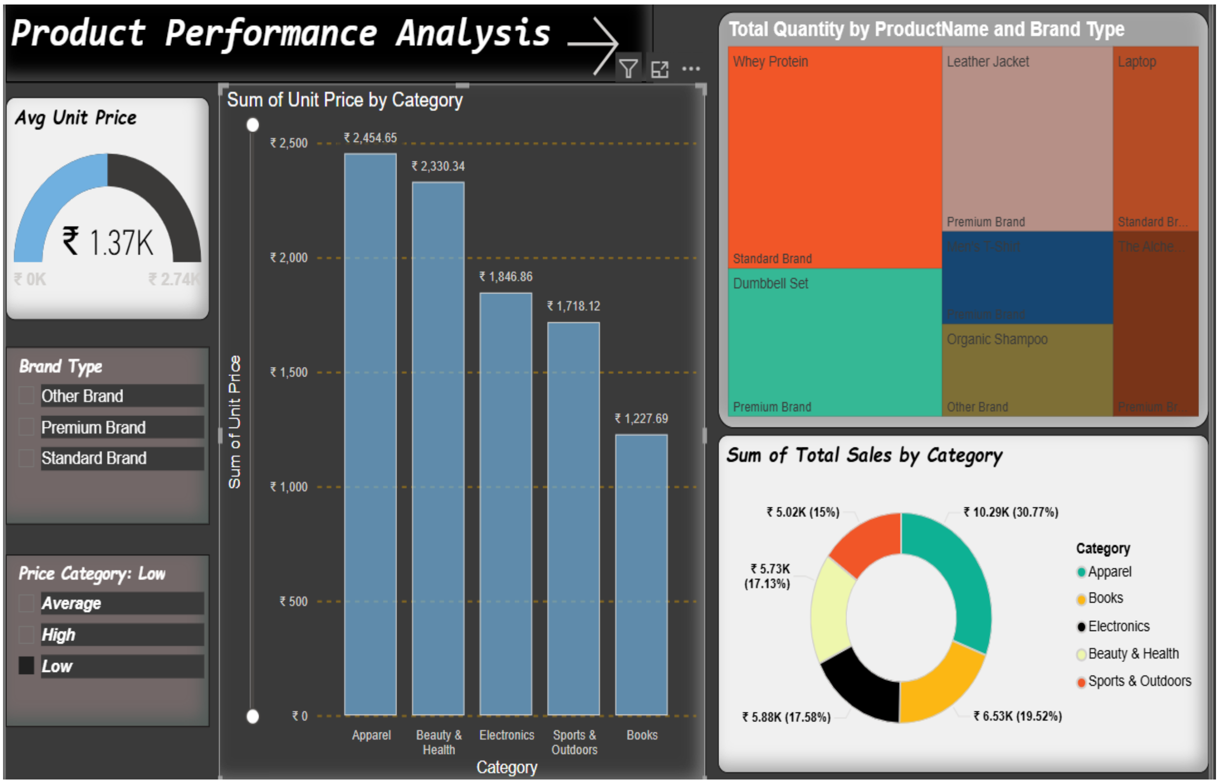
8. Analysis and Visualizations

Dashboard:

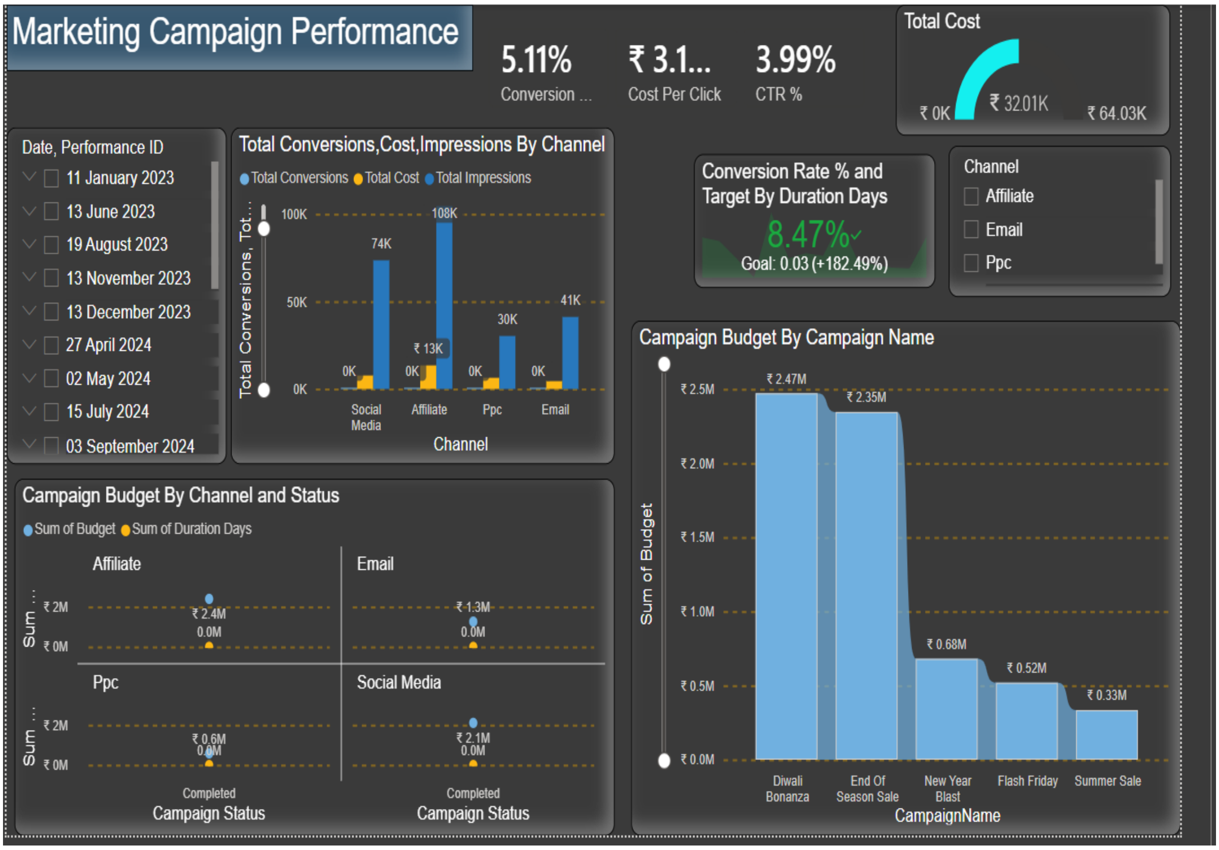
The Power BI dashboard contains 7 visuals that present a complete view of e-commerce business performance. These visuals help analyze sales, customer behavior, product performance, inventory status, order returns, and marketing campaign effectiveness.


KPI and Cards
₹ 279.41865K
Sum of Total Sales
₹ 160.19459K
Sum of Unit Price
10K
Sum of Clicks
515
Sum of Conversions
253K
Sum of Impressions
₹ 3.1754
Cost Per Click
3.99%
CTR %
5.11%
Conversion Rat…
₹ 32.01…
Total Cost
Sum of Budget by EndDate
₹ 496.89756K
Sum of Total Sales and Forecasted Sales by Duration Days
₹ 279.42K!
Goal: ₹ 307.360515K (-9.09%)
Total-Sales
₹ 279.42K
₹ 0K
₹ 558.84K
1K
Total_Orders-Disti…
7K
Sum of Stock Qua…
34
Total Products
214
Processing Orders
199
Cancelled Orders
195
Delivered Orders


Ecommerce Sales and Marketing Analysis
₹ 279.41865K
Sum of Total Sales
1K
Total_Orde…
Order
800
Total Customers
₹ 32.01453K
Total Cost
60
Sum of Quantity
Order Date
01-01-2023
21-06-2026
Country, City
All
Order Status
All
Category
Apparel
Beauty & Health
Brand
BrandA
BrandX
Total Customers by City
Increase
Decrease
Total
Total Customers
City
Surat
Mumbai
Bangalore
Ahmedabad
Chennai
Hyderabad
Delhi
Pune
Jaipur
Kolkata
Total
95
90
84
80
80
80
74
74
73
70
800
Order Status By Total Orders
181 (18.1%)
214 (21.4%)
195 (19.5%)
211 (21.1%)
199 (19.9%)
Order Status
Processing
Pending
Cancelled
Delivered
Shipped
Monthly Sales
Sum of Total Sales
Total Customers
₹ 91K
₹ 90K
₹ 55K
₹ 43K
₹ 80K
₹ 60K
₹ 40K
800
1000
600
Sum of Total Sales
Total Customers
2023
2024
2025
2026
Year

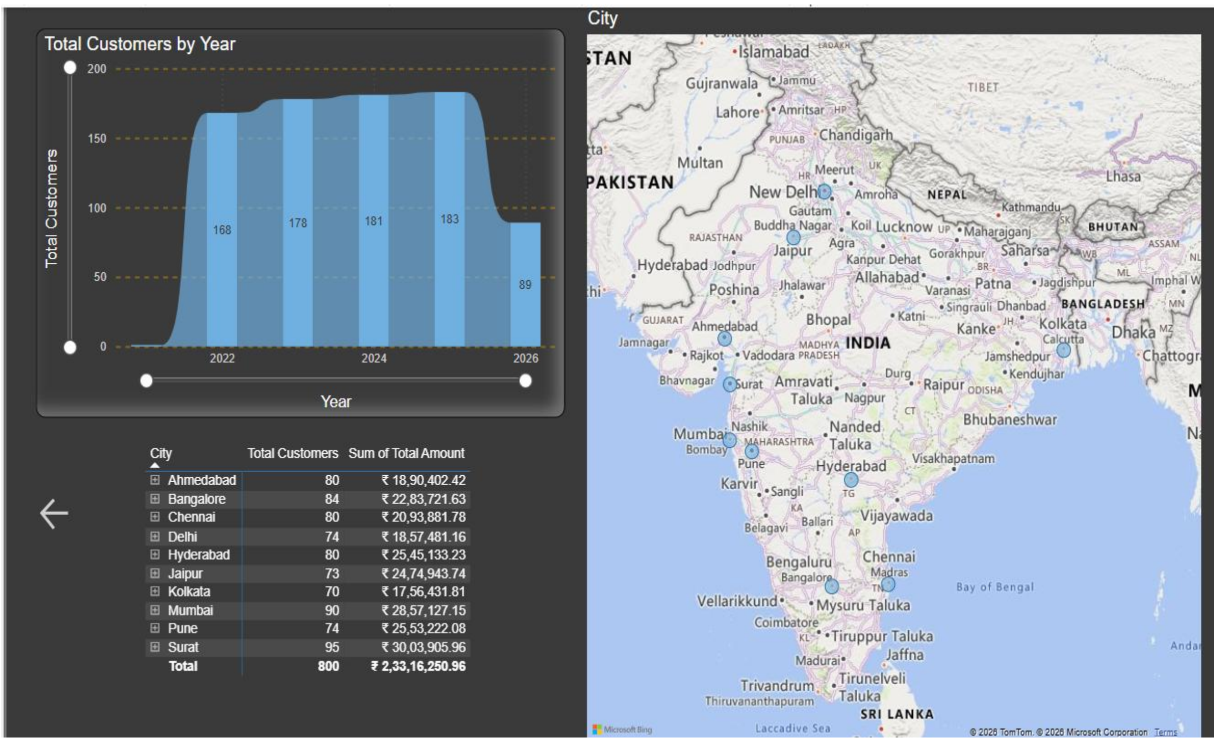


Product Performance Analysis
Avg Unit Price
₹ 1.37K
₹ 0K
₹ 2.74K
Brand Type
Other Brand
Premium Brand
Standard Brand
Price Category: Low
Average
High
Low
Sum of Unit Price by Category
₹ 2,454.65
₹ 2,330.34
₹ 1,846.86
₹ 1,718.12
₹ 1,227.69
Sum of Unit Price
Apparel
Beauty & Health
Electronics
Sports & Outdoors
Books
Category
Total Quantity by ProductName and Brand Type
Whey Protein
Leather Jacket
Laptop
Standard Brand
Premium Brand
Dumbbell Set
Organic Shampoo
Other Brand
Sum of Total Sales by Category
₹ 5.02K (15%)
₹ 10.29K (30.77%)
₹ 5.73K (17.13%)
₹ 5.88K (17.58%)
₹ 6.53K (19.52%)
Category
Apparel
Books
Electronics
Beauty & Health
Sports & Outdoors



Marketing Campaign Performance
5.11%
Conversion ...
₹ 3.1...
Cost Per Click
3.99%
CTR %
Total Cost
₹ 0K
₹ 32.01K
₹ 64.03K
Date, Performance ID
11 January 2023
13 June 2023
19 August 2023
13 November 2023
13 December 2023
27 April 2024
02 May 2024
15 July 2024
03 September 2024
Total Conversions,Cost,Impressions By Channel
Total Conversions
Total Cost
Total Impressions
Social Media
Affiliate
Ppc
Email
Channel
Conversion Rate % and Target By Duration Days
8.47%
Goal: 0.03 (+182.49%)
Channel
Affiliate
Email
Ppc
Campaign Budget By Campaign Name
Sum of Budget
₹ 2.47M
₹ 2.35M
₹ 0.68M
₹ 0.52M
₹ 0.33M
Diwali Bonanza
End Of Season Sale
New Year Blast
Flash Friday
Summer Sale
CampaignName
Campaign Budget By Channel and Status
Sum of Budget
Sum of Duration Days
Affiliate
Email
Ppc
Social Media
Completed
Campaign Status


Warehouse Analysis
Return Orders by Reason
100%
Doesn't Fit 2
Wrong Item 2
Damaged Product 1
Poor Quality 1
50%
Stock By Warehouse
Stock By Warehouse
3K
2K
1K
0K
2.8K
2.6K
2.1K
WH-South
WH-West
WH-North
Warehouse
Stock Status By Warehouse
Avg Stock
Minimum Stock
Max Stock
WH-North
WH-South
WH-West
Avg Stock, Minimum Sto...
1,000
500
0
369
264
305
232
98
183
382
348
347
203
139
407
310
374
219
113
189
Max Stock
Minimum Stock
Avg Stock
High Stock
Low Stock
Medium Stock
Stock Status


Total Customers by Year
Total Customers
200
150
100
50
0
168
178
181
183
89
2022
2024
2026
Year
City
City | Total Customers | Sum of Total Amount
Ahmedabad 80 ₹ 18,90,402.42
Bangalore 84 ₹ 22,83,721.63
Chennai 80 ₹ 20,93,881.78
Delhi 74 ₹ 18,57,481.16
Hyderabad 80 ₹ 25,45,133.23
Jaipur 73 ₹ 24,74,943.74
Kolkata 70 ₹ 17,56,431.81
Mumbai 90 ₹ 28,57,127.15
Pune 74 ₹ 25,53,222.08
Surat 95 ₹ 30,03,905.96
Total 800 ₹ 2,33,16,250.96
PAKISTAN
NEPAL
BHUTAN
BANGLADESH
INDIA
SRI LANKA
TIBET
Bay of Bengal
Islamabad
New Delhi
Jaipur
Ahmedabad
Surat
Mumbai
Pune
Hyderabad
Bengaluru
Chennai
Kolkata
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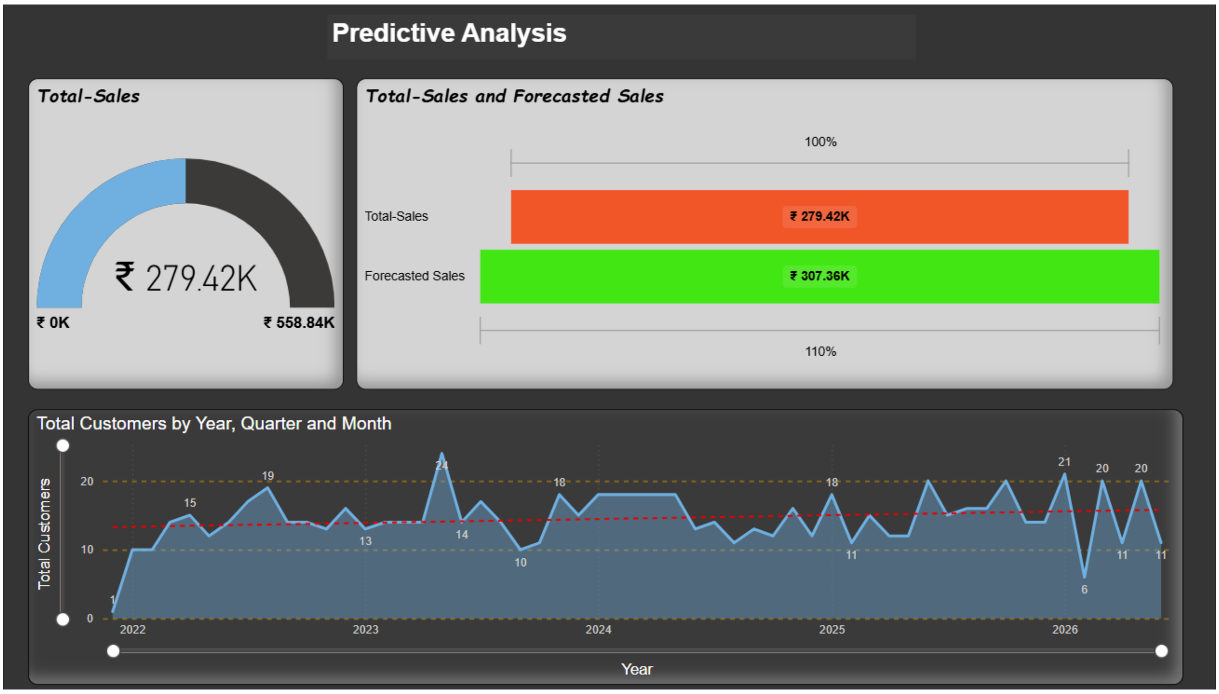

9. Insights & Conclusions

- **Provide the analysis insights:**

Goal: Summarize historical data and current performance.

Key Insights:

- **Total Sales:** ₹279.42K with **1K orders** and **800 customers**.
- **Top Cities:** Surat, Mumbai, Bangalore, and Ahmedabad lead in customer count.
- **Order Status:** Delivered (195) and Processing (214) dominate, showing strong fulfillment efficiency.
- **Product Categories:** Electronics (₹33.9K) and Sports and Outdoor(₹31.6K) contribute most to highest price but low sales.
- **Campaigns:** Affiliate channel generated the highest impressions (108K) and conversions (515).
- **Interpretation:** Your business shows steady growth across major metro cities, with strong product diversity and effective marketing campaigns driving conversions.

Goal: Identify reasons behind performance trends.

Findings:

- **High sales in metro cities** due to better brand visibility and campaign reach.
- **Apparel and Electronics** outperform other categories because of frequent promotions and high consumer demand.
- **Affiliate campaigns** outperform others due to higher budget allocation and longer duration.
- **Return rates** are low, indicating good product quality and customer satisfaction.
- **Root Causes:**
- Effective targeting in high-income urban areas.
- Strong brand positioning for Premium brands (Prestige, Quantum, Stellar).
- Balanced inventory management preventing stockouts.

- **Goal:** Forecast future trends using historical patterns.

Predictions:

- **Sales Growth:** Expected to rise by 10–15% in the next quarter if current campaign performance continues.
- **Customer Expansion:** Likely to reach 1,000+ customers by end of 2026, driven by metro city growth.
- **Top Performing Categories:** Electronics and Apparel will remain dominant, with Beauty & Health showing emerging potential.

Here we forecast future trends using the data.

- Customer Growth: If the trend continues, by 2027 we can expect 950–1000 customers.
- City Expansion: Surat and Mumbai will likely remain leaders, but emerging growth is visible in Bangalore and Hyderabad.
- Seasonality: Sign-ups peak around March–June, suggesting seasonal campaigns or academic cycles.
- Average unit price ₹4.71K and category sales distribution help forecast future revenue.
- High-performing categories: **Home & Kitchen (25%)**, **Books (23%)**, **Beauty & Health (19%)** — these will likely continue driving sales.
- Brand quantity trends (Prestige, Quantum, BrandA) show stable performance, predicting steady product demand.

Goal: Recommend actions to improve future outcomes. **Business Recommendations:**

- **Expand campaigns** in Tier-2 cities (Jaipur, Surat, Ahmedabad) to capture growing demand.

- **Increase budget** for Affiliate and Social Media channels — proven high ROI.
- **Focus on Premium brands** (Prestige, Quantum, Stellar) for higher margins.
- **Introduce loyalty programs** to retain existing customers and boost repeat orders.
- **Enhance inventory forecasting** using Power BI predictive models to avoid overstock or shortages.

Area	Insight	Recommendation
Sales	₹279K total, strong metro performance	Expand Tier-2 city campaigns
Customers	800 total, steady growth	Launch loyalty & referral programs
Products	Apparel & Electronics lead	Diversify into Beauty & Health
Marketing	Affiliate channel most effective	Increase budget allocation
Inventory	Balanced stock levels	Use predictive restocking models

10. Conclusions

The project demonstrates how data analytics can support better decision-making in e-commerce by identifying sales trends, customer behavior, product performance, inventory status, and marketing effectiveness. The Power BI dashboard provides an interactive and user-friendly platform for monitoring business performance. Based on the insights, the business should focus on expanding campaigns in Tier-2 cities, increasing investment in high-performing marketing channels, improving inventory forecasting, and introducing customer retention strategies such as loyalty and referral programs.

THANK YOU